

- 1) [Boxplots for Distribution Comparisons] Construct boxplots to compare the distribution of monthly rainfall across 5 different months of the monsoon season.

Python code:

```
import matplotlib.pyplot as plt
```

```
# Monthly rainfall data (in mm) for 5 monsoon months
```

```
june = [120, 135, 140, 150, 160, 170, 180, 190, 200, 210]
```

```
july = [220, 230, 240, 250, 260, 270, 280, 290, 300, 310]
```

```
august = [180, 190, 200, 210, 220, 230, 240, 250, 260, 270]
```

```
september = [140, 150, 155, 160, 170, 175, 180, 190, 200, 210]
```

```
october = [80, 90, 95, 100, 110, 120, 130, 140, 150, 160]
```

```
# Combine data
```

```
rainfall = [june, july, august, september, october]
```

```
# Labels
```

```
months = ["June", "July", "August", "September", "October"]
```

```
# Create boxplot
```

```
plt.figure(figsize=(8, 6))
```

```
plt.boxplot(rainfall, labels=months, patch_artist=True)
```

```
# Title and labels
```

```
plt.title("Boxplots of Monthly Rainfall During Monsoon Season")
```

```
plt.xlabel("Months")
```

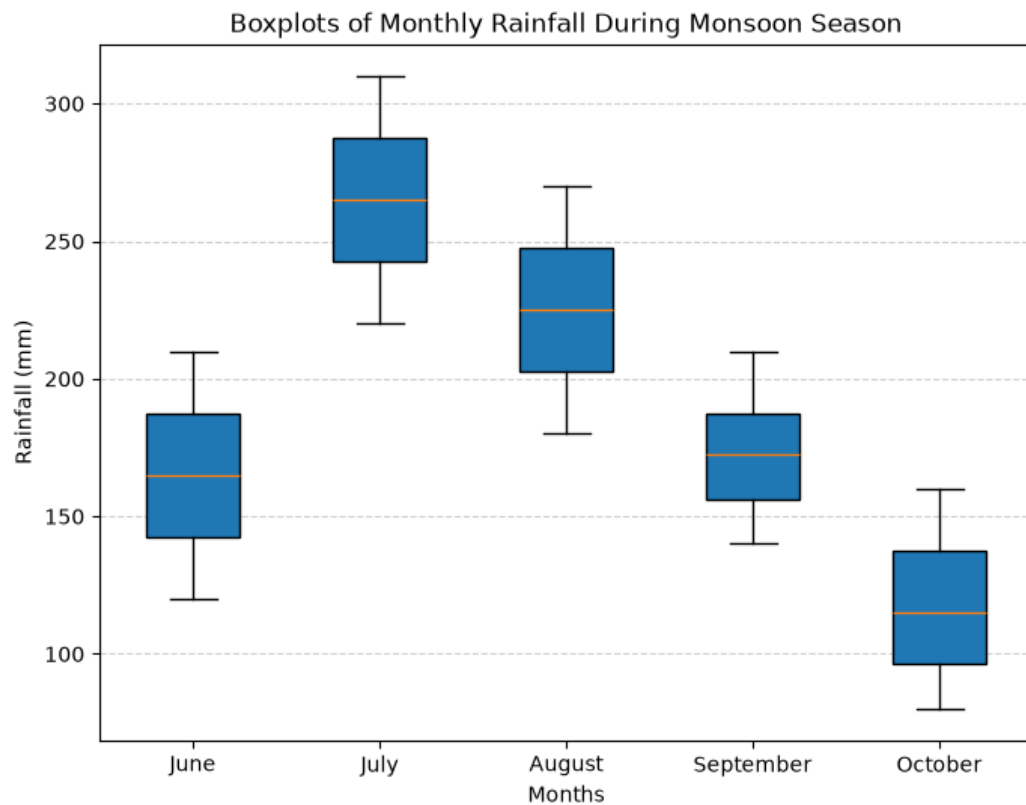
```
plt.ylabel("Rainfall (mm)")
```

```
# Show plot
```

```
plt.grid(axis='y', linestyle='--', alpha=0.6)
```

```
plt.show()
```

output:



- 2) [Amounts] Using data on the number of patients treated by 5 departments in a hospital last month, construct a chart to visualize these amounts with clear axis labels and a title.

Python code:

```
import matplotlib.pyplot as plt
```

```
# Hospital departments
```

```
departments = ["Cardiology", "Neurology", "Orthopedics", "Pediatrics", "General  
Medicine"]
```

```
# Number of patients treated last month
```

```
patients = [320, 210, 280, 350, 420]
```

```
# Create bar chart

plt.figure(figsize=(8, 6))

plt.bar(departments, patients)


# Title and axis labels

plt.title("Number of Patients Treated by Hospital Departments (Last Month)")

plt.xlabel("Hospital Departments")

plt.ylabel("Number of Patients")


# Display values on top of bars

for i, value in enumerate(patients):

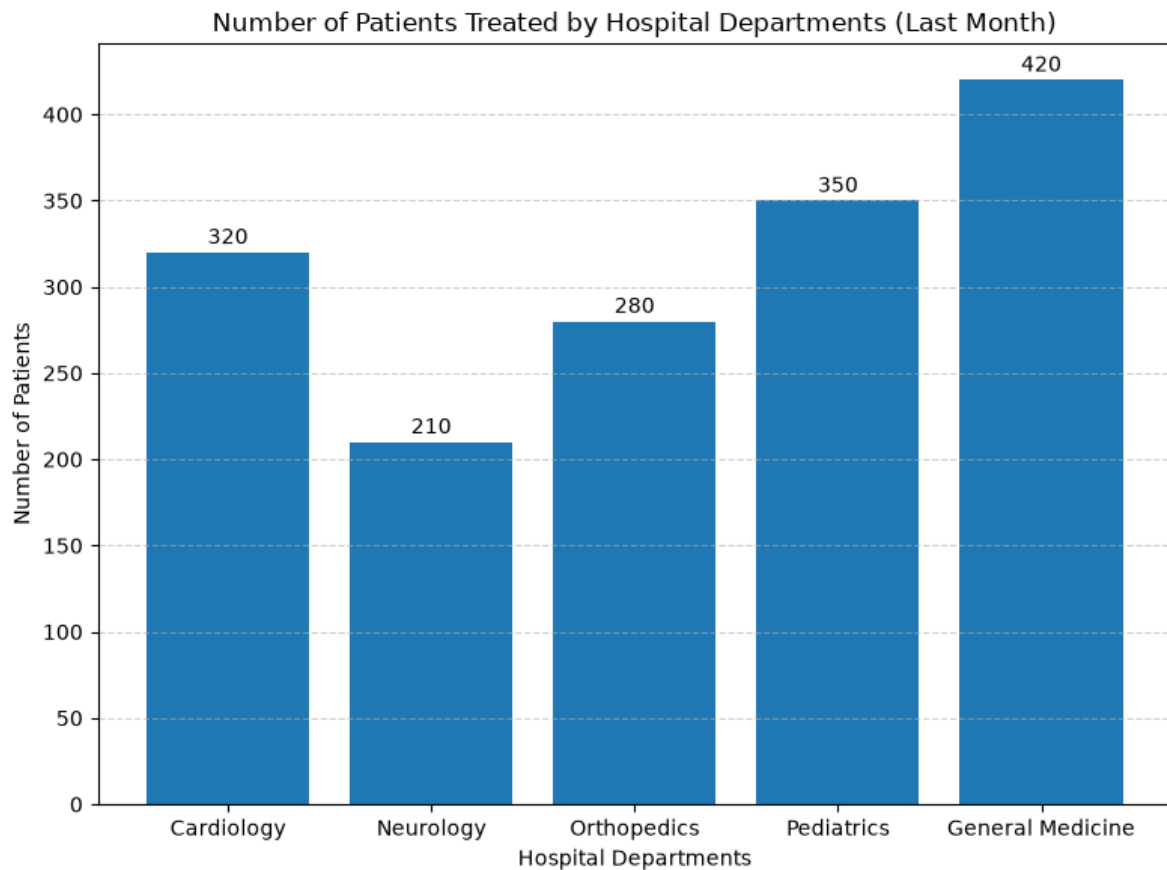
    plt.text(i, value + 5, str(value), ha='center')


plt.grid(axis='y', linestyle='--', alpha=0.6)

plt.tight_layout()

plt.show()

output:
```



3) [Distributions] Given the delivery times (in minutes) of 35 food-delivery orders, construct a chart that best represents the distribution of these times.

Python code:

```
import matplotlib.pyplot as plt
```

```
# Delivery times (in minutes) for 35 food-delivery orders
```

```
delivery_times = [
```

```
    18, 22, 25, 20, 27, 30, 24, 26, 19,
```

```
    35, 28, 21, 23, 29, 32, 31, 26, 27,
```

```
    24, 22, 20, 18, 25, 30, 33, 36, 28,
```

```
    27, 23, 21, 19, 34, 29, 26, 24
```

```
]
```

```
# Create histogram
```

```
plt.figure(figsize=(8, 6))  
plt.hist(delivery_times, bins=6, edgecolor='black')
```

```
# Title and axis labels
```

```
plt.title("Distribution of Food Delivery Times")
```

```
plt.xlabel("Delivery Time (minutes)")
```

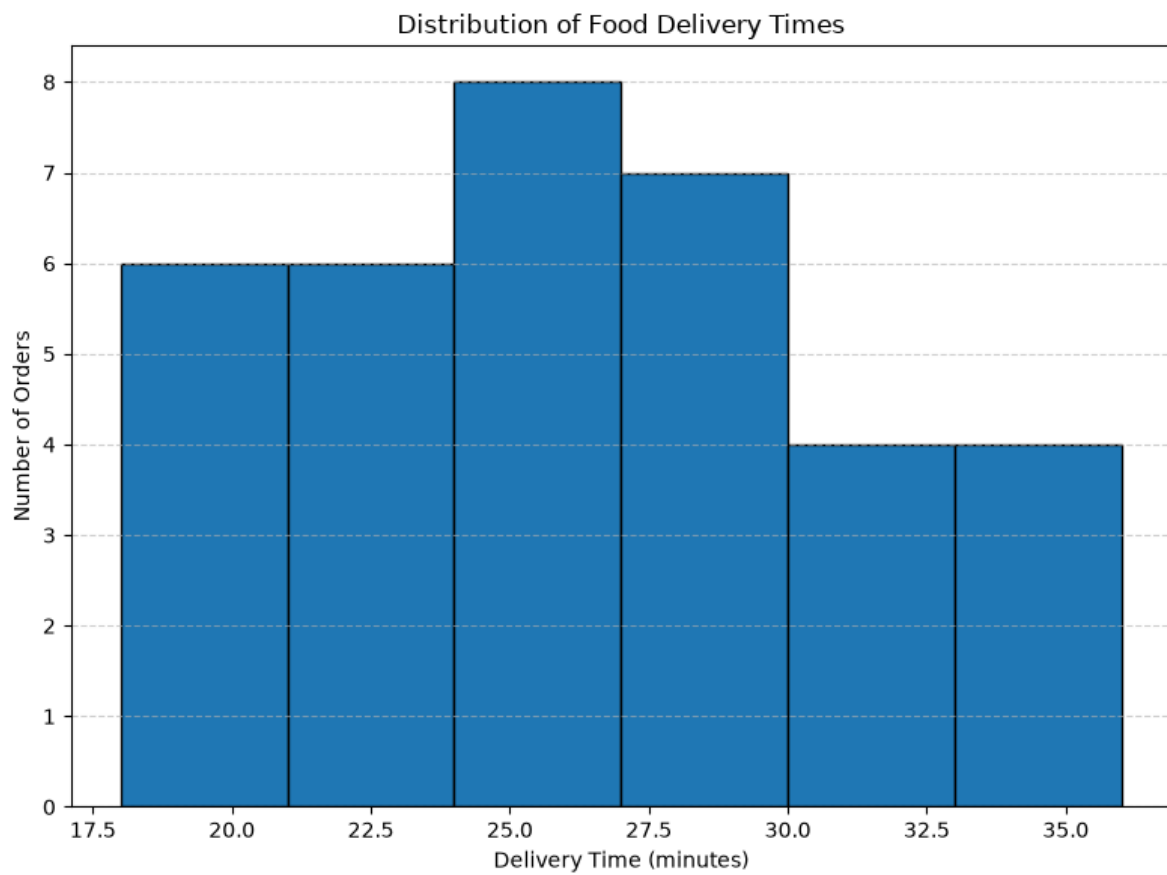
```
plt.ylabel("Number of Orders")
```

```
plt.grid(axis='y', linestyle='--', alpha=0.6)
```

```
plt.tight_layout()
```

```
plt.show()
```

output:



- 4) [Proportions] Using data on the proportion of monthly expenses (%) of a household across 5 categories (rent, food, transport, utilities, savings), construct an appropriate chart.

Python code:

```
import matplotlib.pyplot as plt
```

```
# Expense categories
```

```
categories = ["Rent", "Food", "Transport", "Utilities", "Savings"]
```

```
# Monthly expense proportions (%)
```

```
expenses = [35, 25, 15, 10, 15]
```

```
# Create pie chart
```

```
plt.figure(figsize=(7, 7))
```

```
plt.pie(
```

```
    expenses,
```

```
    labels=categories,
```

```
    autopct="%1.1f%%",
```

```
    startangle=90
```

```
)
```

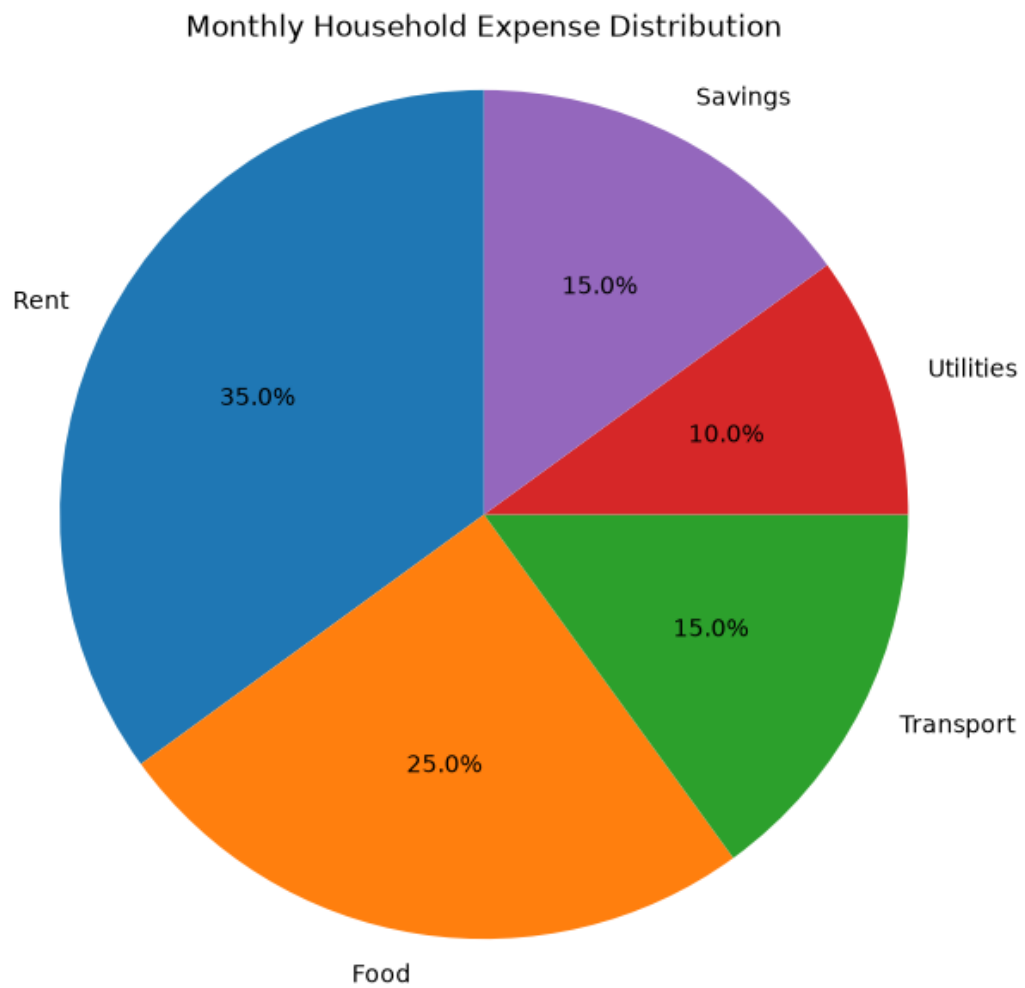
```
# Title
```

```
plt.title("Monthly Household Expense Distribution")
```

```
plt.axis("equal") # Makes the pie chart circular
```

```
plt.show()
```

output:



- 5) [x-y Relationships] Given data on years of work experience and monthly salary (₹) of 22 employees, construct an appropriate chart to explore this x-y relationship.

Python code:

```
import matplotlib.pyplot as plt
```

```
# Years of work experience
```

```
experience = [
```

```
    1, 2, 2, 3, 3, 4, 5, 5, 6, 7, 8,
```

```
    9, 10, 11, 12, 13, 14, 15, 16, 18, 20, 22
```

```
]
```

```
# Monthly salary (₹)

salary = [
    22000, 25000, 26000, 28000, 30000, 32000,
    35000, 36000, 39000, 42000, 45000,
    48000, 52000, 56000, 60000, 65000,
    70000, 76000, 82000, 90000, 100000, 115000
]

# Create scatter plot

plt.figure(figsize=(8, 6))

plt.scatter(experience, salary, marker='o')

# Title and labels

plt.title("Relationship Between Work Experience and Monthly Salary")
plt.xlabel("Years of Work Experience")
plt.ylabel("Monthly Salary (₹)")

plt.grid(True, linestyle='--', alpha=0.6)

plt.tight_layout()

plt.show()

output:
```

